

MD. ASAD-AL-ADIL

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Rajshahi, Bangladesh

RESEARCH PROFILE

Final-year Computer Science and Engineering undergraduate at **Rajshahi University of Engineering & Technology (RUET)** with a **CGPA of 3.93/4.00**, focused on **VLSI**, hardware acceleration, and efficient **AI computing**. My undergraduate thesis, **HeSA-TAPoT**, develops a **boundary-aware heterogeneous systolic architecture** for multiplierless CNN acceleration using **two-term Additive Power-of-Two (APoT) quantization**. I have hands-on experience with **Verilog RTL**, **STM32 embedded systems**, **Pytorch-based CNNs**, and the **Cadence VLSI flow**, with **two conference manuscripts currently under review**. I am seeking a research-oriented Master's opportunity in VLSI, AI accelerators, and energy-efficient computer architecture.

RESEARCH INTERESTS

- VLSI / Digital IC Design
- AI/ML Accelerators
- Energy-Efficient & Parallel Computing
- FPGA/ASIC Acceleration
- Systolic Architectures
- Hardware-Software Co-design

EDUCATION

2022–2027	B.Sc. in Computer Science and Engineering , RUET, Bangladesh	CGPA: 3.93/4.00
2021	Higher Secondary Certificate (HSC)	GPA: 5.00/5.00
2019	Secondary School Certificate (SSC)	GPA: 5.00/5.00

HONORS & LEADERSHIP

Student of the Year Award – 2nd Academic Year	Year-2023
Class Representative Appreciation – 3rd Academic Year,	Year-2024

UNDERGRADUATE THESIS

HeSA-TAPoT: A Boundary-Aware Heterogeneous Systolic Array Architecture for Multiplierless CNN Acceleration Using Two-Term Additive Power-of-Two Quantization Ongoing

- Developing a **boundary-aware heterogeneous systolic architecture** in **Verilog HDL** for efficient multi-channel CNN acceleration.
- Using **two-term APoT quantization** to replace weight multiplications with two shift operations and an addition, enabling a multiplierless convolution datapath with heterogeneous processing elements.
- Designing and verifying convolution, accumulation, pooling, bias, and requantization stages while studying throughput, resource use, numerical precision, and energy-efficiency trade-offs.

CONFERENCE MANUSCRIPTS UNDER REVIEW

ICCIT 2026 – Manuscript on Software-Level Two-Term Shift-Based CNN Quantization Submitted / Under Review

Develops and evaluates a software-level **two-term shift representation** in which CNN weight multiplication is approximated using **dual shift operations plus addition**, providing the algorithmic basis for multiplierless hardware acceleration.

ICECE 2026 (BUET) – Manuscript on Knowledge-Distilled CNN Inference on MCU Submitted / Under Review

Uses **knowledge distillation** to obtain a compact CNN for microcontroller inference; a preprocessed **image array is transmitted to the MCU**, which performs digit prediction and returns a **single predicted digit**.

SELECTED ENGINEERING PROJECTS

Handwritten Digit Recognition using STM32 and UART

[GitHub](#)

- Developed a TensorFlow CNN for handwritten-digit recognition using **standard and custom datasets**, achieving **98.2% classification accuracy**; classification was performed with TensorFlow's `predict()` function.
- Built a Python-based handwritten-input acquisition and preprocessing pipeline and transmitted the **predicted digit through UART** to an **STM32 microcontroller**.
- Programmed the STM32 in embedded C to receive the classification result and display it on a **7-segment display** through custom digital interfacing.

Medical Charges Predictor using Machine Learning

Hugging Face

Built and deployed a Gradient-Boosting model for healthcare-cost estimation using preprocessing, feature engineering, hyperparameter tuning, cross-validation, and an interactive inference interface.

Additional Software Projects

MyExpense: Django/SQLite personal-finance web application (GitHub); Expense Tracker: Java/Android application for persistent expense recording and monthly visualization (GitHub).

RESEARCH & TECHNICAL EXPERIENCE

VLSI Design Trainee, Neural Semiconductor Limited, Dhaka

23 Feb 2026 – 6 Mar 2026

- Gained hands-on exposure to the digital VLSI design flow using Cadence, including RTL design, functional verification, physical-layout concepts, and fabrication-oriented constraints.
- Studied DRC/LVS verification and the transition from digital design to silicon implementation.

TECHNICAL SKILLS

HDL / Architecture: Verilog HDL, RTL Design, Digital Logic, Systolic / PE-based Architecture

Programming: C, C++, Python, Java

VLSI / EDA: Cadence, KiCad, Logisim Evolution

Embedded Systems: STM32F103C8, Embedded/Bare-Metal C, UART, Arduino, Proteus, Keil μVision

AI / ML: TensorFlow, PyTorch, Knowledge Distillation, Model Compression, Scikit-learn, NumPy, Pandas

Development: Git/GitHub, VS Code, Matplotlib, Plotly

ADDITIONAL EXPERIENCE

Teaching Assistant & Academic Script Evaluator, Private Educational Organization

Sep 2022 – Present

- Support academic instruction and have evaluated more than 13,000 school-level STEM examination scripts using structured rubrics and written performance feedback.
- Provide one-to-one private tutoring and personalized academic support in Physics, Mathematics, and IT-related subjects for students requiring individualized guidance.

INTERESTS

Strength Training & Fitness: Maintaining a consistent strength-training routine for approximately six years; placed 5th in the Mr. Rajshahi Competition 2024.

ACADEMIC REFERENCES

Dr. Md. Nazrul Islam Mondal

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